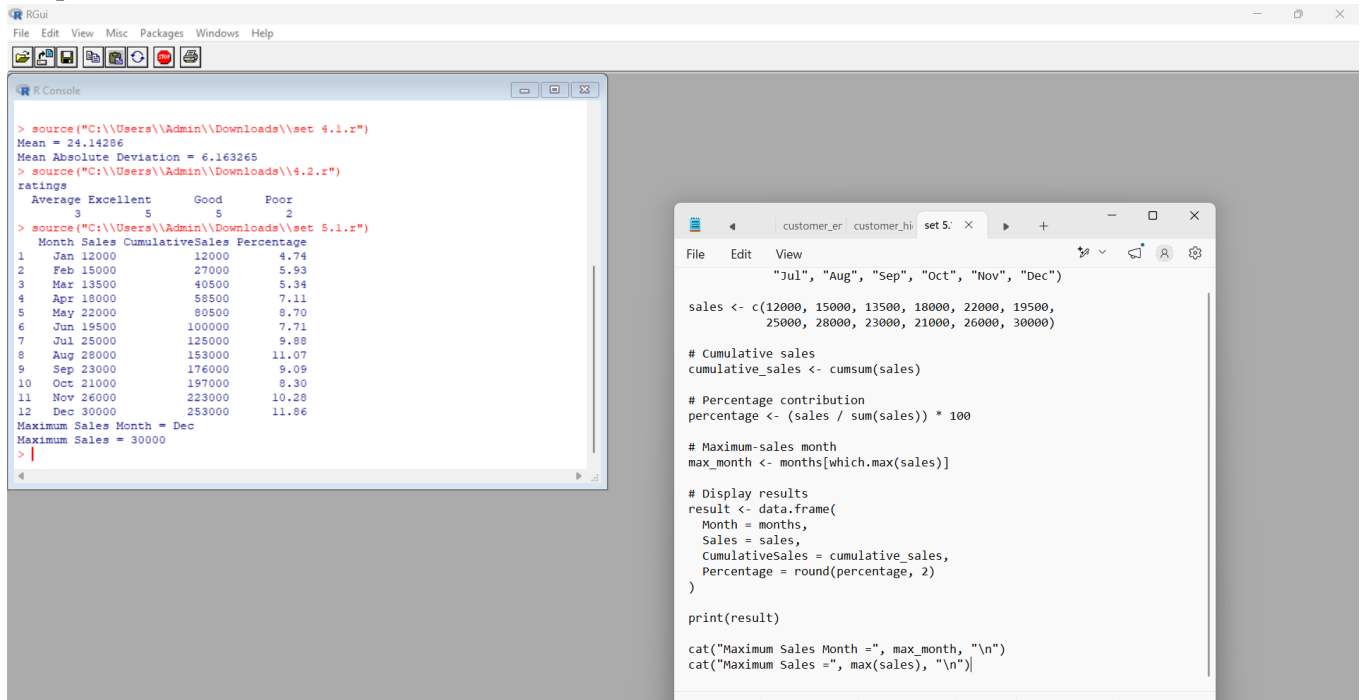


SET 5

1. R: Create 12 months of sales; calculate cumulative sales, percentage contribution and maximum-sales month.

Output:



The screenshot shows the RGui interface. The R Console on the left displays the following output:

```
> source("C:\\Users\\Admin\\Downloads\\set 4.1.r")
Mean = 24.14286
Mean Absolute Deviation = 6.163265
> source("C:\\Users\\Admin\\Downloads\\set 4.2.r")
ratings
  Average Excellent   Good   Poor
        3         5     5     2
> source("C:\\Users\\Admin\\Downloads\\set 5.1.r")
  Month Sales CumulativeSales Percentage
1 Jan 12000      12000      4.74
2 Feb 15000     27000      5.93
3 Mar 13500     40500      5.34
4 Apr 18000     58500      7.11
5 May 22000     80500      8.70
6 Jun 19500    100000      7.71
7 Jul 25000    125000      9.88
8 Aug 28000    153000    11.07
9 Sep 23000    176000      9.09
10 Oct 21000   197000      8.30
11 Nov 26000   223000    10.28
12 Dec 30000   253000    11.86
Maximum Sales Month = Dec
Maximum Sales = 30000
> |
```

The script editor on the right shows the following code:

```
customer_er customer_hi set 5.1.r
File Edit View
"Jul", "Aug", "Sep", "Oct", "Nov", "Dec")

sales <- c(12000, 15000, 13500, 18000, 22000, 19500,
          25000, 28000, 23000, 21000, 26000, 30000)

# Cumulative sales
cumulative_sales <- cumsum(sales)

# Percentage contribution
percentage <- (sales / sum(sales)) * 100

# Maximum-sales month
max_month <- months[which.max(sales)]

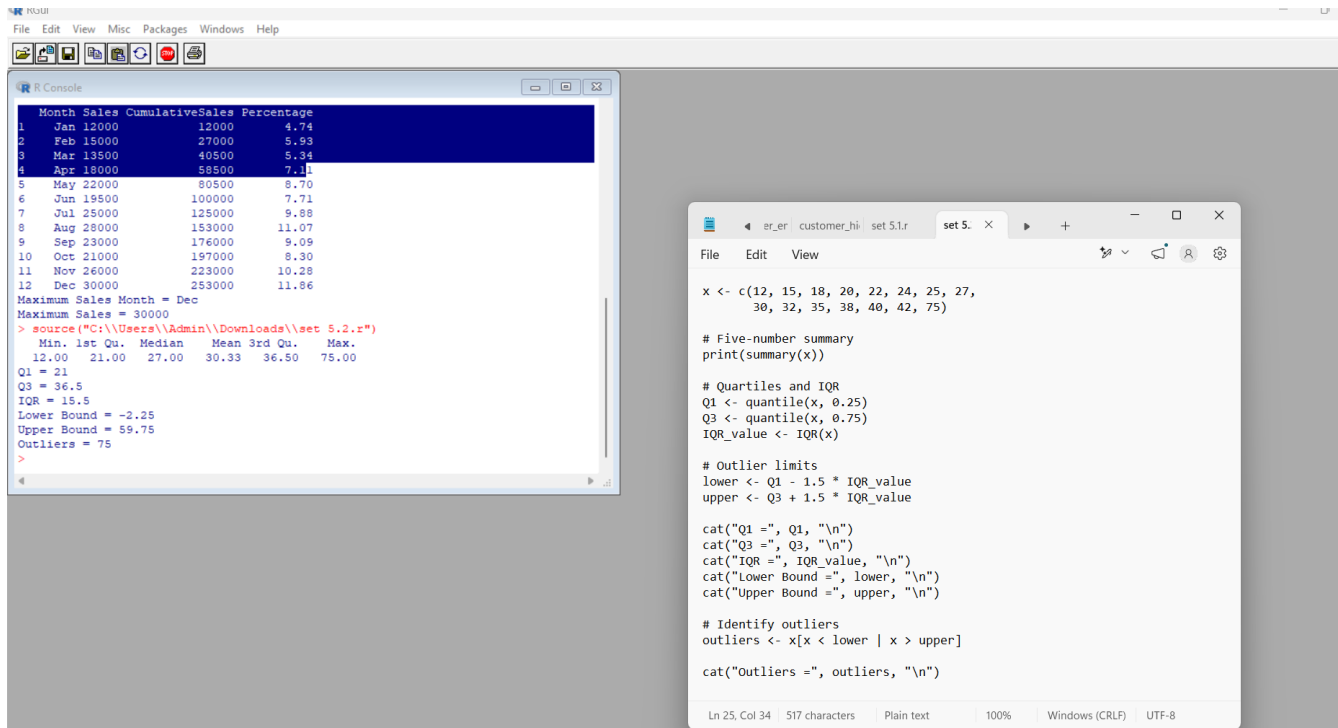
# Display results
result <- data.frame(
  Month = months,
  Sales = sales,
  CumulativeSales = cumulative_sales,
  Percentage = round(percentage, 2)
)

print(result)

cat("Maximum Sales Month =", max_month, "\n")
cat("Maximum Sales =", max(sales), "\n")
```

2. R: Find the five-number summary for 15 observations and identify outliers using $1.5 \times \text{IQR}$.

Output:



3. WEKA: Create loan-applicant ARFF data (Income, Age, Employment, LoanStatus); apply J48 and visualize.

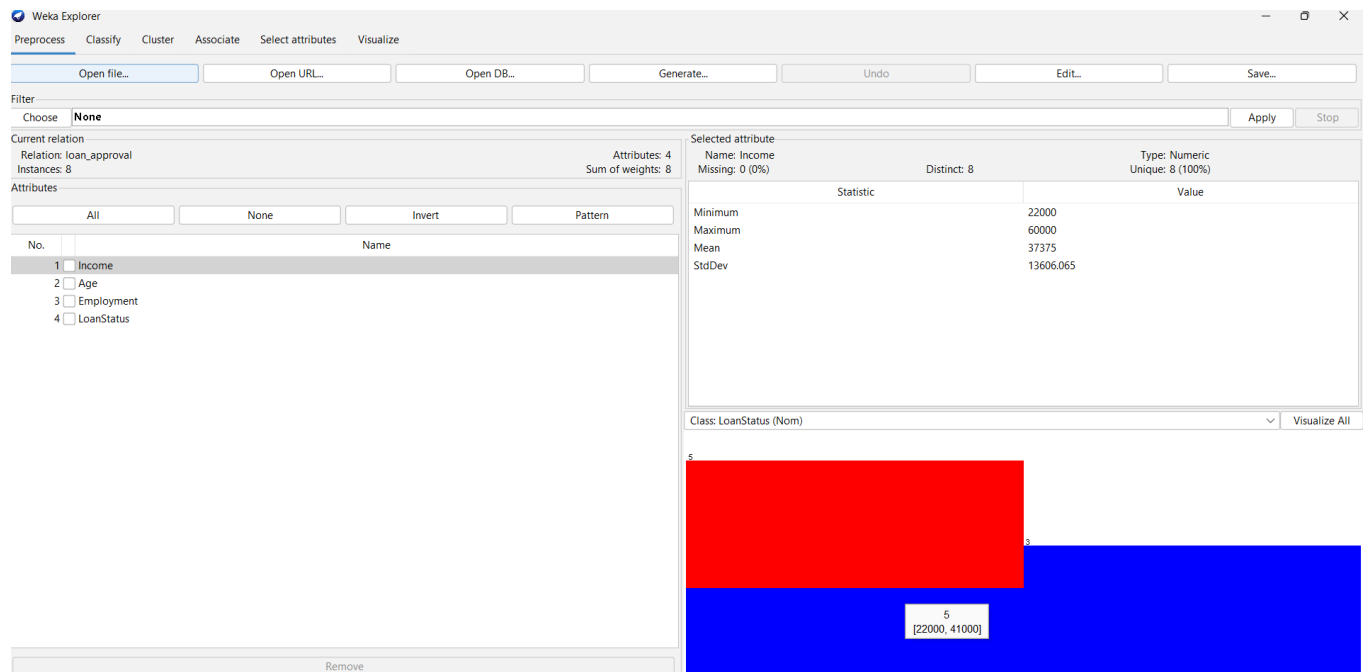
Dataset: Loan Approval

```

@relation loan_approval
@attribute Income numeric
@attribute Age numeric
@attribute Employment {Employed,SelfEmployed,Unemployed}
@attribute LoanStatus {Approved,Rejected}
@data
25000,23,Employed,Rejected
32000,28,Employed,Approved
45000,35,Employed,Approved
28000,31,SelfEmployed,Rejected
60000,42,SelfEmployed,Approved
22000,25,Unemployed,Rejected
52000,38,Employed,Approved
35000,29,SelfEmployed,Approved

```

Output:



4. WEKA: Apply SMO to the loan dataset and compare accuracy and confusion matrix with J48.

Dataset: Loan Approval – SMO

Use the same Loan Approval ARFF dataset from Set 5 Q3.

Output:

